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**DISK FORENSICS AND TIMELINE RECONSTRUCTION USING
AUTOPSY AND THE SLEUTH KIT**

A BLUE TEAM PERSPECTIVE ON DIGITAL FORENSICS AND INCIDENT RESPONSE

BY

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CHAPTER ONE: INTRODUCTION

1.1 Background

Growing reliance on digital systems has increased both the frequency and impact of cybersecurity incidents, making reliable methods for collecting and analysing digital evidence essential. Digital forensics identifies, preserves, and examines electronic evidence in a manner that maintains its integrity for investigative and legal use. Disk forensics, one of its most important branches, recovers traces of user activity.

Timeline reconstruction correlates timestamps recovered from multiple artifact sources, the file system, the Windows Registry, browser databases, and system logs into a single coherent narrative of events. This capability is central to Blue Team operations, where defenders must reconstruct an attacker's actions accurately and quickly to scope, contain, and learn from an incident. This project uses Autopsy, built on The Sleuth Kit, together with AccessData FTK Imager, to demonstrate this process in practice.

1.2 Problem Statement

Organisations routinely face insider threats, unauthorised data access, data exfiltration via removable media, and malware persistence, often compounded by attempts to delete files or clear logs. Without a structured forensic methodology and the right tools, an investigator may fail to recover critical evidence or produce findings that cannot withstand scrutiny. This project addresses the need for a repeatable approach that recovers digital artifacts, correlates them into an accurate timeline, preserves evidence integrity, and produces outputs that are directly useful to a Blue Team seeking to detect similar incidents in future.

1.3 Aim of the Project

The aim of this project is to perform a practical disk forensic investigation and reconstruct an accurate timeline of user and system activity using Autopsy and The Sleuth Kit, within a controlled laboratory environment, while demonstrating how the resulting evidence supports Blue Team detection and incident response objectives.

1.4 Objectives

- Design and simulate a realistic forensic investigation scenario involving an insider threat and data exfiltration.
- Acquire a forensically sound disk image of a Windows 10 virtual machine using AccessData FTK Imager.
- Preserve evidence integrity through cryptographic hashing and a documented chain of custody.
- Analyse the acquired image using Autopsy and The Sleuth Kit to recover digital artifacts.
- Recover deleted files and identify evidence of anti-forensic activity.
- Reconstruct a chronological timeline of user and system activity from the recovered artifacts.

- Map the recovered evidence to relevant MITRE ATT&CK techniques from a Blue Team perspective.
- Present findings, analysis, and recommendations in a professional forensic project report.

1.5 Scope of the Project

This project is limited to disk forensic analysis of a single Windows 10 virtual machine running on VMware Workstation Pro, hosted on a Windows 11 physical machine. It focuses on file system, Registry, and web browser artifacts, and timeline reconstruction using Autopsy and The Sleuth Kit. Network forensics, live memory forensics, mobile forensics, and malware reverse engineering are out of scope. Any script referenced in the simulated scenario was functionally inert and used solely to generate realistic forensic artifacts for analysis.

1.6 Significance of the Study

This project demonstrates, in a hands-on and reproducible manner, how open-source forensic tools support both digital forensic investigation and Blue Team detection engineering. Rather than treating forensics as purely reactive, it connects each recovered artifact to a corresponding detection opportunity, illustrating the practical link between forensics, incident response, and proactive defence. The methodology can serve as a reference for students and small organisations building in-house forensic readiness using freely available tools.

1.7 Organisation of the Report

This report is organised into seven chapters. Chapter One introduces the project's background, problem, aim, objectives, and scope. Chapter Two reviews relevant literature and tools. Chapter Three describes the methodology and laboratory setup. Chapter Four presents the practical implementation. Chapter Five presents the results and findings. Chapter Six analyses the findings and presents Blue Team recommendations. Chapter Seven concludes the report and suggests future work.

CHAPTER TWO: LITERATURE AND TOOL REVIEW

2.1 Digital and Disk Forensics

Digital forensics is the application of scientific and analytical techniques to the identification, preservation, collection, examination, and presentation of digital evidence in a manner acceptable for investigative and legal proceedings (Casey, 2011). Disk forensics, its most established branch, focuses on recovering and examining information stored on hard disks, solid state drives, and removable media (Carrier, 2005), including deleted files, file system metadata, Registry data, and browser artifacts. Because most file systems mark deleted space as available for reuse rather than erasing it immediately, deleted data frequently remains recoverable until overwritten, making disk forensics a powerful technique for recovering evidence of user activity even after deliberate deletion.

2.2 Timeline Reconstruction

Timeline reconstruction arranges forensic artifacts according to their timestamps to determine the chronological sequence of events on a system. Because a single user action typically generates multiple related artifacts across the disk, correlating them by time builds a more complete picture than any single artifact could provide alone. This is especially valuable in incident response, where establishing an accurate sequence of attacker actions is essential to scoping a compromise.

2.3 Core Tools: The Sleuth Kit, Autopsy, and FTK Imager

The Sleuth Kit (TSK) is an open-source collection of command line tools for analysing disk images and file systems, including FAT, NTFS, and Unix/Linux systems, providing utilities for listing files (including deleted entries), extracting content, and generating file system timelines. It forms the underlying engine of Autopsy, an open-source graphical platform that simplifies investigation through integrated modules for deleted file recovery, browser and Registry analysis, keyword search, hash lookup. Autopsy operates on a forensic image without modifying the original evidence, preserving integrity throughout analysis.

AccessData FTK Imager was used for acquisition. It creates forensic images in formats such as raw (dd) and Expert Witness Format (E01), automatically computing MD5 and SHA1 hashes during imaging to allow later verification that the image has not been altered.

VMware Workstation Pro, a type 2 hypervisor, was used to create the isolated Windows 10 victim machine on which the scenario was simulated, allowing repeatable, risk-free execution from a preserved clean baseline.

2.4 Forensic Standards and Frameworks

This project was guided by recognised digital forensic principles, including the NIST Guide to Integrating Forensic Techniques into Incident Response (NIST SP 800-86), the ACPO Good Practice Guide for Digital Evidence, general chain of custody principles requiring a documented, unbroken record of evidence handling, and verification of evidence integrity through cryptographic hashing at acquisition and analysis.

2.5 Comparative Overview of Forensic Tools

Table 2.1 compares Autopsy, FTK Imager, and representative commercial alternatives to contextualise the tool selection made for this project.

Table 2.1: Comparison of Common Digital Forensic Tools

Tool	Type	Primary Use	Cost
The Sleuth Kit / Autopsy	Open source	Disk image analysis, artifact recovery, timeline reconstruction	Free
AccessData FTK Imager	Free utility	Forensic disk acquisition and hashing	Free
EnCase Forensic	Commercial	Full forensic investigation suite with advanced reporting	Licensed
X-Ways Forensics	Commercial	Lightweight, high-performance forensic analysis	Licensed

Autopsy and FTK Imager were selected because they are freely available, widely used in academic and professional settings, and sufficiently capable of demonstrating disk forensic and timeline reconstruction techniques without requiring a commercial licence.

CHAPTER THREE: METHODOLOGY

3.1 Research Design and Approach

This project adopted an experimental, laboratory-based approach: a realistic security incident was deliberately simulated within an isolated virtual environment, after which a complete forensic investigation was carried out against the resulting evidence. Knowing the exact sequence and timing of events in advance, the ground truth allowed the accuracy of the Autopsy-generated reconstruction to be independently validated, mirroring how controlled red team versus blue team exercises are used in professional training.

3.2 Laboratory Setup and Architecture

The laboratory environment consisted of a single physical host running a virtualisation platform, on top of which an isolated victim virtual machine was created. Table 3.1 summarises the configuration used.

Table 3.1: Hardware and Software Specification of the Laboratory Environment

Component	Specification
Host operating system	Windows 11
Virtualisation platform	VMware Workstation Pro
Victim virtual machine OS	Windows 10
Forensic acquisition tool	AccessData FTK Imager, version 3.3.0
Forensic analysis platform	Autopsy, version 4.23.1 (built on The Sleuth Kit)
Forensic image format	Expert Witness Format (E01)
Hashing algorithms	MD5 and SHA-1 / SHA-256

The laboratory followed a linear evidentiary workflow: the Windows 11 host ran VMware Workstation Pro, which hosted the isolated Windows 10 victim machine on which simulated activity was performed. After the scenario, the virtual disk was acquired with FTK Imager into an E01 image, which was imported into Autopsy for ingest processing and timeline generation.

3.3 Forensic Scenario Design

To generate realistic, forensically meaningful artifacts, a scenario was designed around a common incident type: an insider threat involving exfiltration of a sensitive file to removable media, download of a suspicious script, establishment of Registry persistence, and a subsequent attempt to cover these actions through file deletion and history clearing. This scenario exercises a broad range of artifact categories relevant to a Blue Team investigation while remaining safe and repeatable in a student laboratory setting. No genuinely malicious code was used at any stage.

3.4 Procedure

The overall procedure spanned scenario simulation, evidence acquisition, and forensic analysis:

1. Took a clean baseline snapshot of the virtual machine prior to any simulated activity.
2. Simulated connection of a removable USB storage device to the virtual machine.
3. Created a sensitive document and copied it to the simulated removable storage device.
4. Downloaded a functionally inert script disguised as a legitimate file.
5. Created a Windows Registry Run key entry to simulate persistence.
6. Deleted the sensitive document and emptied the Recycle Bin to simulate anti-forensic behaviour.
7. Cleared a portion of the command history to further simulate covering tracks.
8. Shut down the virtual machine and located its virtual disk file on the host.
9. Acquired a forensic image of the disk using FTK Imager in Expert Witness Format, recording the hash values.
10. Created a new case in Autopsy and added the forensic image as a data source.
11. Configured and executed the relevant Autopsy ingest modules against the image.
12. Examined recovered artifacts and analysis results, and generated the reconstructed timeline.
13. Correlated the recovered evidence into a chronological narrative and compared it against the ground truth.

3.5 Evidence Acquisition and Preservation

The victim virtual machine's disk file was first copied into a dedicated evidence directory, separate from working files, to avoid accidental modification of the original. FTK Imager was then used to create a forensic image in Expert Witness Format, during which it automatically computed cryptographic hash values for the source and destination data. These hashes were recorded as part of the chain of custody, providing a means of verifying that the image analysed in Autopsy was identical to the original evidence at acquisition.

CHAPTER FOUR: IMPLEMENTATION

4.1 Scenario Creation and Evidence Generation

Implementation began with preparing the Windows 10 victim virtual machine in VMware Workstation Pro. A clean snapshot, labelled Clean State, was taken before any simulated activity, allowing the exercise to be repeated from a known baseline. The insider threat scenario from Chapter Three was then carried out entirely within the isolated virtual machine, with each action performed in a controlled order and its real-world time of execution recorded as ground truth for later validation. Table 4.1 presents this ground truth log.

Table 4.1: Ground Truth Log of Simulated Incident Actions

Step	Simulated Action	Approx. Time	Expected Artifact
1	USB storage device connected	6:00 PM	Registry USBSTOR, Devices Attached
2	Sensitive document copied to removable media	6:05 PM	File system timeline entry
3	Suspicious script downloaded	6:09 PM	Web download record, browser history
4	Registry Run key persistence created	6:15 PM	Registry Run Programs artifact
5	Document deleted, Recycle Bin emptied	6:23 PM	Deleted file / MFT, Recycle Bin
6	Command history cleared	6:29 PM	Gap in command history logs
7	Virtual machine shut down	6:31 PM	System shutdown event

Exact timestamps recorded during the laboratory session should be inserted above and later compared against timestamps recovered by Autopsy in Chapter Five to validate the reconstructed timeline.

4.2 Disk Imaging Using FTK Imager

After the scenario, the virtual machine was shut down and its virtual disk file copied into a dedicated evidence folder. FTK Imager 3.3.0 was used to image this disk, with source type set to Image File, destination format set to E01, and case information entered before imaging began. Figure 4.1 shows the Drive/Verify results.



Drive/Image Verify Results	
Name	ForensicImage.E01
Sector count	83886080
MD5 Hash	
Computed hash	aca4ed3fc6992c5e4d7712afa4baf2a4
Stored verification hash	aca4ed3fc6992c5e4d7712afa4baf2a4
Report Hash	aca4ed3fc6992c5e4d7712afa4baf2a4
Verify result	Match
SHA1 Hash	
Computed hash	ef9206a6c292d5f8438bcca5680c29c3581cd59a
Stored verification hash	ef9206a6c292d5f8438bcca5680c29c3581cd59a
Report Hash	ef9206a6c292d5f8438bcca5680c29c3581cd59a
Verify result	Match
Bad Sector List	
Bad sector(s)	No bad sectors found

Figure 4.1: FTK Imager Create Disk Image dialog showing the selected source and destination image settings.

Upon completion, FTK Imager produced the E01 image with an accompanying log containing the MD5 and SHA1 hash values, summarised in Table 4.2, which constitute the chain of custody record.

Table 4.2: Forensic Image Acquisition Details

Item	Detail
Source	Windows 10 virtual machine disk (.vmdk)
Destination image format	Expert Witness Format (E01)
Imaging tool	AccessData FTK Imager 3.3.0
MD5 hash	aca4ed3fc6992c5e4d7712afa4baf2a4
SHA1 hash	ef9206a6c292d5f8438bcca5680c29c3581cd59a
Verification result	verified

4.3 Importing the Image and Configuring Ingest

A new case, Case01, was created in Autopsy 4.23.1, and the E01 image was added as a data source. The relevant ingest modules were then selected and executed, each contributing a different category of recovered artifact.

Table 4.3: Autopsy Ingest Modules Enabled

Ingest Module	Purpose
Recent Activity	Recovers recently accessed files, programs, and OS usage artifacts
Hash Lookup	Compares file hashes against known sets to flag notable/ignorable files
File Type Identification	Identifies true file types independent of extension
Extension Mismatch Detector	Flags files whose extension does not match their true type
Registry Analysis	Extracts and interprets Registry hive contents, including persistence keys and USB history
Keyword Search	Indexes disk content for targeted searches
Timeline	Builds the event data used by Autopsy's Timeline tool

Ingest processing was allowed to run to completion before any artifact examination began, ensuring the full set of recoverable evidence was available for analysis.

4.4 Timeline Generation

Once ingest completed, the Timeline tool was launched and zoomed into the date/time range corresponding to the simulated scenario. Each artifact of interest, described in Chapter Five, was cross-referenced against its position in the timeline using Autopsy's View File in

Timeline function, confirming the chronological relationship between artifacts directly within the tool.

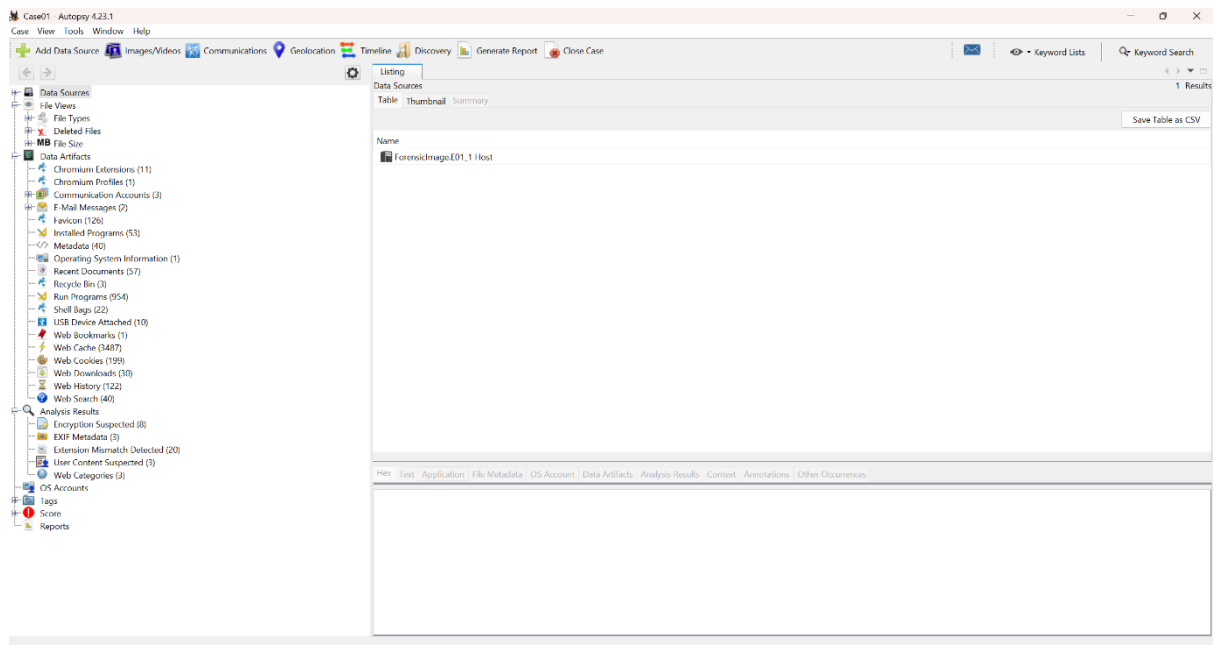


Figure 4.5: Autopsy window with forensic image.

CHAPTER FIVE: RESULTS AND FINDINGS

5.1 Overview of Recovered Artifacts

Ingest processing produced a substantial volume of recovered data spanning browser activity, installed software, document activity, removable media history, and Registry persistence indicators. Table 5.1 summarises the principal categories recovered.

Table 5.1: Summary of Data Artifacts Recovered by Autopsy

Artifact Category	Entries Recovered
Chromium Extensions	11
Chromium Profiles	1
Communication Accounts	3
E-Mail Messages	2
Favicon	126
Installed Programs	53
Metadata	40
Operating System Information	1
Recent Documents	57
Recycle Bin	3
Run Programs	954
Shell Bags	22
USB Device Attached	10
Web Bookmarks	1
Web Cache	3,487
Web Cookies	199
Web Downloads	30
Web History	122
Web Search	40

Run Programs returned the largest number of entries (954), reflecting both legitimate startup entries and the single simulated persistence entry. The USB Device Attached, Web Downloads, Recent Documents, and Recycle Bin categories are of particular relevance to the scenario and are discussed individually below.

5.2 Operating System and Installed Programs

The Operating System Information artifact (Figure 5.1) confirmed the Windows version, host name, and owner details, serving as a baseline check that the correct image had been analysed. Autopsy also recovered 53 installed program entries (Figure 5.2), including the web browser used for the simulated download — a routine step establishing the system's baseline software capability.

Header	Test	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Result: 1 of 1	Result								
Operating System Information									
Type									Source(s)
Name									Recent Activity
Program Name									Recent Activity
Processor Architecture									Recent Activity
Temporary Files Directory									Recent Activity
Path									Recent Activity
Product ID									Recent Activity
Owner									Recent Activity
Source File Path									
Artifact ID									

Figure 5.1: Operating System Information artifact recovered by Autopsy, confirming the identity and configuration of the analysed system.

Listing									
Installed Programs									53 Results
Table	Thumbnail	Summary							Save Table as CSV
Source Name	S	C	O	Program Name	Date/Time	Data Source			
SOFTWARE			0	Office 16 Click-to-Run Extensibility Component 64-bit ...	2026-07-28 21:15:08 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Office 16 Click-to-Run Licensing Component v.16.0.17...	2026-07-28 21:10:57 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Microsoft Office Professional Plus 2024 - en-us v.16.0...	2026-07-28 21:10:46 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Microsoft Update Health Tools v.3.74.0.0	2026-07-28 21:06:05 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Update for Windows 10 for x64-based Systems (KB50...	2026-07-28 21:05:53 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	WinRAR 7.22 (64-bit) v.7.22.0	2026-07-28 20:56:50 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Python 3.14.5 Add to Path (64-bit) v.3.14.5150.0	2026-07-17 04:51:02 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Python 3.14.5 Td/Tk Support (64-bit) v.3.14.5150.0	2026-07-17 04:51:01 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Python 3.14.5 Documentation (64-bit) v.3.14.5150.0	2026-07-17 04:50:41 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Python 3.14.5 Test Suite (64-bit) v.3.14.5150.0	2026-07-17 04:50:37 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Python 3.14.5 Standard Library (64-bit) v.3.14.5150.0	2026-07-17 04:50:26 GMT-04:00	ForensicImage.E01			
SOFTWARE			0	Python 3.14.5 Development Libraries (64-bit) v.3.14.5...	2026-07-17 04:50:16 GMT-04:00	ForensicImage.E01			

Figure 5.2: Installed Programs artifact listing software identified on the victim virtual machine.

5.3 Web Browser Artifacts

A substantial number of browser artifacts were recovered, including 122 Web History entries, 3,487 Web Cache entries, 199 Web Cookies, 40 Web Search entries, and 30 Web Downloads. The Web Downloads category (Figure 5.3) identified the previously downloaded script with its recorded timestamp, confirming that the download left a directly recoverable trace independent of the file system.

Web Downloads									
Table	Thumbnail	Summary							30 Results
									Save Table as CSV
Source Name	S	C	O	Path	URL				
History			0	C:\Users\theamanor\Downloads\Rich-Dad-Poor-Dad...	https://drive.usercontent.google.com/uc?id=087Or...				
History			0	C:\Users\theamanor\Downloads\Rich-Dad-Poor-Dad...	https://drive.usercontent.google.com/download?id=...				
History			0	C:\Users\theamanor\Downloads\th.webp	https://th.bing.com/th?id=OIF.38CK6pAQH%2f22...				
History			0		https://tse3.mm.bing.net/th?id/OIP.9FD1tq7IKZ0u...				
History			0		https://tse1.mm.bing.net/th?id/OIP.0JbZk5mDd4ei...				
History			0		https://cdn.britannica.com/10/250610-050-BC5CCI...				
History			0	C:\Users\theamanor\Downloads\images.jpg	https://encrypted-tbn0.gstatic.com/images?q=tbn...				
History			0	C:\Users\theamanor\Downloads\accounting_system_e...	https://samplefile.com/samples/download/archive/...				
History			0	C:\Users\theamanor\Downloads\nohemi-font-family.zip	https://befonts.com/download?post_id=397514&pc...				
History			0	C:\Users\theamanor\Downloads\nohemi-font-family (...)	https://befonts.com/download?post_id=397514&pc...				
History			0		https://th.bing.com/th?id/OIP.U53o5mMj-5p09YsPI...				

Figure 5.3: Web Downloads artifact showing the recovered download record and associated timestamp for the simulated script.

5.4 Recent Documents and Recycle Bin

The Recent Documents artifact (Figure 5.4) returned 57 entries, including the sensitive document accessed prior to deletion - evidence that a file was used even where it has since been removed. The Recycle Bin artifact (Figure 5.5) returned only 3 entries, since it was deliberately emptied during the scenario; the deleted document instead required examination of the broader Deleted Files view (Section 5.7).

Source Name	S	C	O	Path	Date Accessed	Data Source	
All Tasks.Ink				No preferred path found	2026-07-17 00:48:08 GMT-04:00	ForensicImage.EC	
The Internet.Ink				No preferred path found	2026-07-17 00:18:25 GMT-04:00	ForensicImage.EC	
ms-settingsnetwork.Ink				No preferred path found	2026-07-17 00:37:34 GMT-04:00	ForensicImage.EC	
ms-settingsnetwork-ethernet.Ink				No preferred path found	2026-07-17 00:18:25 GMT-04:00	ForensicImage.EC	
No preferred path found.Ink				No preferred path found	0000-00-00 00:00:00	ForensicImage.EC	
Desktop.LNK				C:\Users\theamanor\Desktop	2026-07-28 13:47:41 GMT-04:00	ForensicImage.EC	
KPAN (F) (2).Ink				F:\	2026-07-24 21:05:41 GMT-04:00	ForensicImage.EC	
KPAN (F).Ink				F:\	2026-07-24 21:05:41 GMT-04:00	ForensicImage.EC	
m1.Ink				C:\Users\theamanor\Downloads\m1.jpg	2026-07-28 14:35:04 GMT-04:00	ForensicImage.EC	
m3.Ink				C:\Users\theamanor\Downloads\m3.jpg	2026-07-28 14:35:16 GMT-04:00	ForensicImage.EC	
M4.Ink				C:\Users\theamanor\Downloads\M4.jpg	2026-07-28 14:36:05 GMT-04:00	ForensicImage.EC	

Figure 5.4: Recent Documents artifact showing recently accessed files, including the simulated sensitive document.

Source Name	S	C	O	Path	Time Deleted	Username	Data Source
\$R2IWOE6.zip				C:\Users\theamanor\Downloads\nohemi-font-family (1).zip	2026-07-28 14:39:39 GMT-04:00		ForensicImage.EC
\$R8QZHA2.txt				C:\Users\theamanor\Desktop\Secret Documents\finance_report.txt	2026-07-28 14:42:26 GMT-04:00		ForensicImage.EC
\$R9TC3CA.docx				C:\Users\theamanor\Desktop\Secret Documents\Company VPN Password.docx	2026-07-28 14:42:20 GMT-04:00		ForensicImage.EC

Figure 5.5: Recycle Bin artifact showing entries recovered despite deliberate emptying during scenario simulation.

5.5 USB Device and Persistence Analysis

The USB Device Attached artifact (Figure 5.6) returned 10 Registry-recovered entries, one matching the removable device connected during simulation; its timestamp was cross-referenced against the ground truth log (Table 4.1), providing direct evidence relevant to the exfiltration stage. The Run Programs artifact (Figure 5.7) returned 954 entries; a targeted keyword search isolated the specific Registry Run key created during simulation, confirming a persistence mechanism referencing the downloaded script and supporting the indicator of compromise discussed in Section 6.2.

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
SYSTEM			0	2026-07-28 16:51:29 GMT-04:00		ROOT_HUB	5&2891968b&0	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:29 GMT-04:00		ROOT_HUB20	5&36a4b5d6&0	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:29 GMT-04:00		ROOT_HUB30	5&11106705&0&0	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:30 GMT-04:00	VMware, Inc.	Virtual USB Hub	6&39d724fe&0&7	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:30 GMT-04:00	VMware, Inc.	Virtual USB Hub	6&39d724fe&0&8	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:30 GMT-04:00	VMware, Inc.	Virtual Mouse	6&39d724fe&0&5	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:30 GMT-04:00	VMware, Inc.	Virtual Mouse	7&8bcbfcc2&0&0000	ForensicImage.EC
SYSTEM			0	2026-07-28 16:51:33 GMT-04:00	VMware, Inc.	Virtual Mouse	7&8bcbfcc2&0&0001	ForensicImage.EC
SYSTEM			0	2026-07-28 14:38:24 GMT-04:00	JMicron Technology Corp. / JMicron USA Technology C.	JMS578 SATA 6Gb/s	AA202508130218	ForensicImage.EC
SYSTEM			0	2026-07-17 00:35:59 GMT-04:00		VID_349C&PID_0418	20240418000000	ForensicImage.EC

5.6 Deleted File Recovery

The Deleted Files view (Figure 5.8) was used to search the file system directly for the sensitive document removed from the Recycle Bin. Because NTFS deletion typically marks

the Master File Table entry as reusable rather than erasing the data, the document remained recoverable at imaging time, and its content and metadata matched the file created during simulation, illustrating the practical value of disk forensics against simple anti-forensic deletion.

Name	Keyword Preview	Location	Modified Time
Web Downloads Artifact	/Users/theamanor/Desktop/Secret Documents/Mee...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/...	2026-07-28 17:42:19 GMT-04:00
Recent Documents Artifact	heamanor\Desktop\Secret Documents\Path ID : 264...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/A...	2026-07-28 16:54:13 GMT-04:00
Web Downloads Artifact	/Users/theamanor/Desktop/Secret Documents/emp...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/...	2026-07-28 17:37:20 GMT-04:00
Secret Documents.Ink	dll,-21769SECRET~1\Secret Documents\C\Users\the...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/A...	0000-00-00 00:00:00
Secret Documents.Ink	Simulated Attack\Secret Documents\^Secret Docum...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/A...	0000-00-00 00:00:00
UsrClass.dat.LOG2	shbin SECRET~1\Secret Documents\mk 9[5C4F28B5-F	/img_ForensicImage.E01/vol_vol6/Users/theamanor/A...	2026-07-16 23:05:46 GMT-04:00
Recycle Bin Artifact	heamanor\Desktop\Secret Documents\finance_repo...	/img_ForensicImage.E01/vol_vol6/\$Recycle.Bin/S-1-5...	2026-07-24 21:05:01 GMT-04:00
Recent Documents Artifact	F:\Simulated Attack\Secret Documents\Path ID : -1D...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/A...	0000-00-00 00:00:00
Recycle Bin Artifact	heamanor\Desktop\Secret Documents\Company V...	/img_ForensicImage.E01/vol_vol6/\$Recycle.Bin/S-1-5...	2026-07-28 13:47:42 GMT-04:00
No preferred path found.Ink	\$F:\Simulated Attack\Secret Documents\desktop-m...	/img_ForensicImage.E01/vol_vol6/Users/theamanor/A...	0000-00-00 00:00:00

Figure 5.8: Deleted Files view in Autopsy showing recovery of the sensitive document following its deletion and removal from the Recycle Bin.

5.7 Analysis Results

Autopsy's Analysis Results category flagged additional files based on automated heuristics, summarised in Table 5.2.

Table 5.2: Summary of Analysis Results Flagged by Autopsy

Analysis Result Category	Entries Flagged
Encryption Suspected	8
EXIF Metadata	3
Extension Mismatch Detected	20
User Content Suspected	3
Web Categories	3

The Extension Mismatch Detected category (20 entries) is of particular interest, as this heuristic flags files whose content does not match their extension, a common technique for disguising malicious files. Each entry was reviewed to confirm whether it represented a genuine anomaly or a routine false positive.

5.8 Timeline Reconstruction

The Timeline tool correlated all discussed artifacts into a single chronological view, zoomed into the incident window (Figure 5.9), in which the device connection, file copy, script download, persistence creation, deletion, and history clearing appear in sequence.

Figure 5.9: Reconstructed timeline view in Autopsy showing the chronological sequence of simulated scenario events.

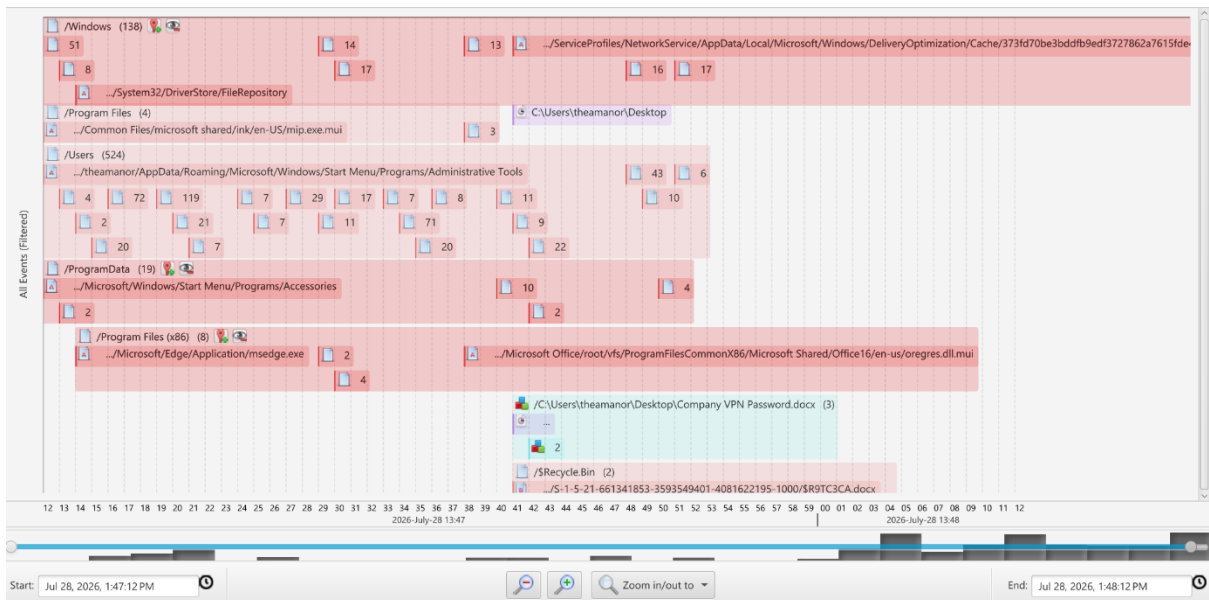


Table 5.3 presents the resulting attack narrative with corresponding MITRE ATT&CK techniques. Recorded times should be populated from the investigator's own analysis and compared against Table 4.1.

Table 5.3: Reconstructed Timeline of Events Mapped to MITRE ATT&CK

Time	Event	Artifact Source	MITRE ATT&CK Technique
6:00 PM	USB device connected	Devices Attached (Registry)	T1091 – Replication Through Removable Media
6:05 PM	Document copied to removable media	File System Timeline	T1005 – Data from Local System
6:09 PM	Script downloaded	Web Downloads	T1105 – Ingress Tool Transfer
6:15 PM	Registry Run key persistence created	Run Programs (Registry)	T1547.001 – Registry Run Keys / Startup Folder
6:23 PM	Document deleted, Recycle Bin emptied	Deleted Files / Recycle Bin	T1070.004 – File Deletion
6:29 PM	Command history cleared	Gap in command history logs	T1070.003 – Clear Command History

The close correspondence between the ground truth log and the Autopsy-reconstructed timeline validates the accuracy of the reconstruction process and demonstrates Autopsy's ability to correctly sequence events drawn from multiple independent artifact sources.

CHAPTER SIX: ANALYSIS AND RECOMMENDATIONS

6.1 Interpretation of Findings

The investigation successfully reconstructed the full sequence of simulated activity from the acquired image. The findings show that considerable evidence survives on a compromised system even after basic anti-forensic actions such as deletion, Recycle Bin emptying, and history clearing, the recovery of the deleted document, despite its removal from the Recycle Bin, illustrates that deletion at the OS level does not equate to destruction at the storage level until the space is overwritten.

The correspondence between the ground truth log (Table 4.1) and the reconstructed timeline (Table 5.3) confirms that Autopsy, built on The Sleuth Kit, can accurately correlate artifacts from disparate sources, the file system, Registry, and browser storage, into a single coherent, chronologically ordered narrative, a capability central to the value disk forensics provides to incident response and Blue Team operations.

6.2 Mapping to the MITRE ATT&CK Framework

Mapping recovered evidence to MITRE ATT&CK expresses forensic findings in a standardised vocabulary directly usable by Blue Team analysts for detection engineering and threat intelligence. Table 6.1 consolidates the techniques identified.

Table 6.1: Mapping of Recovered Artifacts to MITRE ATT&CK Techniques

Recovered Artifact	ATT&CK Technique	Tactic
USB device connection history	T1091	Initial Access / Lateral Movement
File copied to removable media	T1005	Collection
Script download via browser	T1105	Command and Control
Registry Run key persistence	T1547.001	Persistence
Deleted sensitive document	T1070.004	Defense Evasion
Cleared command history	T1070.003	Defense Evasion

This mapping shows that the scenario, despite its simplicity, exercised techniques spanning the Collection, Command and Control, Persistence, and Defense Evasion tactics, providing a realistic and pedagogically useful basis for the exercise.

6.3 Blue Team Implications

Each artifact identified corresponds to a specific, real-time detection opportunity that a properly configured monitoring capability could catch as an incident unfolds, rather than requiring post-hoc disk forensics. For example, a Registry Run key referencing a script in a user-writable Downloads folder is a well-understood persistence pattern detectable via Sysmon Event ID 13 (Registry modification) correlated with Event ID 1 (process creation); unauthorised removable storage connections are similarly detectable via Windows Event ID 6416 or equivalent EDR telemetry. This illustrates a broader principle: forensics and real-

time detection engineering are complementary, and forensic findings should routinely inform monitoring improvements.

6.4 Recommendations

- Deploy endpoint monitoring (e.g., Sysmon or a commercial EDR product) to log process creation, Registry modification, and file system events in real time.
- Enable centralised logging and forward security-relevant events to a SIEM platform for timely detection and correlation.
- Implement device control policies to restrict or monitor unauthorised removable storage.
- Encrypt sensitive files at rest and apply data loss prevention controls to reduce exfiltration risk.
- Develop detection rules (e.g., in Sigma format) targeting known persistence techniques such as Registry Run key modification.
- Conduct regular forensic readiness assessments to ensure logging and acquisition capability supports timely investigation.
- Provide regular security awareness training on data handling and the risks of removable media and untrusted downloads.
- Maintain secure, tested backups to reduce the operational impact of data loss.
- Adopt standardised acquisition and chain of custody procedures, consistent with NIST SP 800-86, for future investigations.

6.5 Limitations of the Study

The investigation was restricted to disk-based artifacts from a single virtual machine and did not incorporate network traffic analysis, live memory forensics, or endpoint telemetry available in a production environment. The simulated scenario, while representative, was simplified for a controlled academic exercise and does not capture the full complexity of a real-world intrusion. All timestamps and hash values referenced should be populated with values obtained during the investigator's own laboratory session.

CHAPTER SEVEN: CONCLUSION

7.1 Summary and Achievement of Objectives

This project set out to perform a practical disk forensic investigation and reconstruct an accurate timeline of activity using Autopsy and The Sleuth Kit, from a Blue Team perspective. A realistic insider threat scenario, removable media exfiltration, a suspicious download, Registry persistence, and anti-forensic activity, was simulated on an isolated Windows 10 virtual machine. A forensic image was acquired with FTK Imager in Expert Witness Format, with hashing preserving evidence integrity, and analysed in Autopsy.

The investigation recovered a substantial range of artifacts, including browser history, downloads, recent documents, Recycle Bin contents, USB history, installed applications, and Registry persistence entries. Deleted files were recovered despite deliberate deletion and Recycle Bin emptying, demonstrating file system resilience against simple anti-forensic techniques. Timeline analysis reconstructed the simulated activities in accurate chronological order, validated against the recorded ground truth, and the evidence was mapped to MITRE ATT&CK techniques and translated into Blue Team recommendations. On this basis, each objective set out in Section 1.4 was achieved.

7.2 Contribution to Blue Team Practice

Beyond demonstrating core disk forensic technique, this project connected each recovered artifact to a corresponding real-time detection opportunity, illustrating the practical relationship between reactive digital forensics and proactive Blue Team detection engineering, reinforcing that forensic findings should routinely inform improvements to logging, monitoring, and detection capability.

7.3 Future Work

- Extend the investigation to network forensics, correlating disk-based evidence with captured traffic.
- Incorporate live memory (RAM) forensics to capture volatile evidence not present on disk.
- Automate correlation of Autopsy findings with a SIEM platform to demonstrate end-to-end detection and response.
- Develop and test a working Sigma detection rule against a simulated SOC environment for the persistence technique identified.
- Repeat the exercise using a super-timeline tool, such as log2timeline and Plaso, to compare artifact correlation depth against Autopsy alone.

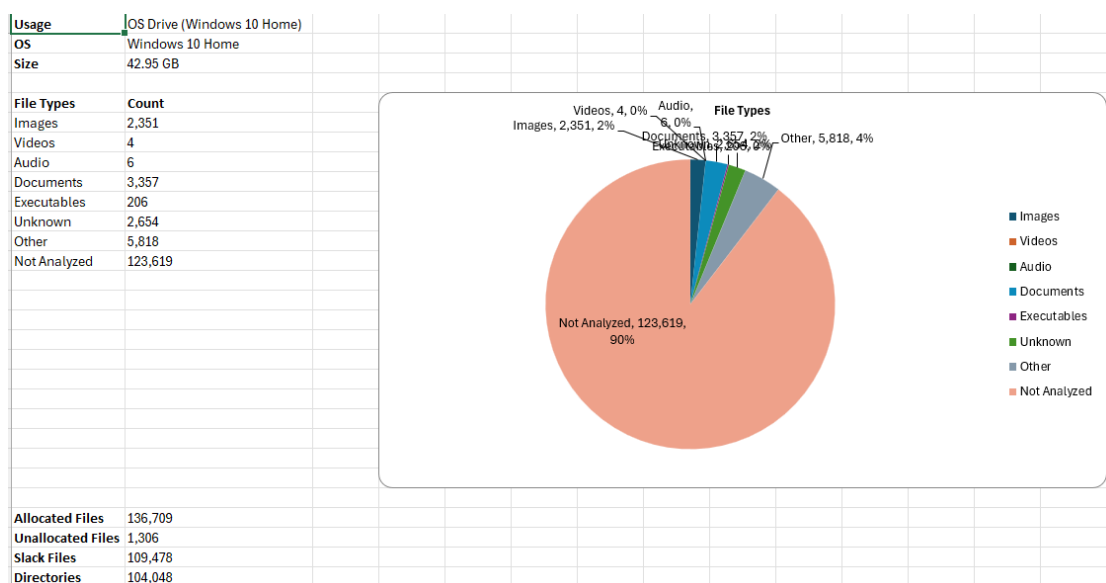
In conclusion, this project provided valuable practical experience in digital forensic investigation, encompassing evidence acquisition, artifact recovery, timeline reconstruction, and evidence-based reporting and reinforced the critical role disk forensics and timeline reconstruction play in supporting modern cybersecurity operations and Blue Team investigations.

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APPENDICES

Appendix A: Full Autopsy case report export.



Appendix B: FTK Imager acquisition log, including recorded hash values.

```
Created By AccessData® FTK® Imager 3.3.0.5

Case Information:
Acquired using: ADI3.3.0.5
Case Number: 001
Evidence Number: 001
Unique Description:
Examiner: Prince Desmond Addo Amanor
Notes:

-----
|
Information for C:\Users\amano\Desktop\Evidence\ForensicImage:

Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 5,221
Heads: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 83,886,080
[Physical Drive Information]
Drive Interface Type: lsilogic
[Image]
Image Type: VMWare Virtual Disk
Source data size: 40960 MB
Sector count: 83886080
[Computed Hashes]
MD5 checksum: aca4ed3fc6992c5e4d7712afa4baf2a4
SHA1 checksum: ef9206a6c292d5f8438bcca5680c29c3581cd59a
```

```
Image Information:
Acquisition started: Tue Jul 28 19:00:53 2026
Acquisition finished: Tue Jul 28 19:08:48 2026
Segment list:
C:\Users\amano\Desktop\Evidence\ForensicImage.E01
C:\Users\amano\Desktop\Evidence\ForensicImage.E02
C:\Users\amano\Desktop\Evidence\ForensicImage.E03
C:\Users\amano\Desktop\Evidence\ForensicImage.E04
C:\Users\amano\Desktop\Evidence\ForensicImage.E05
C:\Users\amano\Desktop\Evidence\ForensicImage.E06
C:\Users\amano\Desktop\Evidence\ForensicImage.E07

Image Verification Results:
Verification started: Tue Jul 28 19:08:48 2026
Verification finished: Tue Jul 28 19:13:52 2026
MD5 checksum: aca4ed3fc6992c5e4d7712afa4baf2a4 : verified
SHA1 checksum: ef9206a6c292d5f8438bcca5680c29c3581cd59a : verified
```